

Notes: Pástor, Stambaugh & Taylor (JFE, 2015)

Scale and skill in active management

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1 Big picture

- **Question.** Do actively managed mutual funds face decreasing returns to scale, and how should we measure skill when scale depresses performance?
- **Main result.** The strongest evidence of decreasing returns is at the industry level: as the active mutual fund industry grows, a fund’s ability to outperform passive benchmarks declines.
- **Fund-level result.** Bias-free estimates also point toward decreasing fund-level returns, but they are not statistically significant.
- **Skill result.** Active management skill rises over time, but industry growth offsets the improvement.
- **Lifecycle result.** Fund performance deteriorates over a typical fund’s lifetime, a pattern explained by industry-level decreasing returns to scale.

2 Incremental contribution

- The paper separates own-fund scale from industry-wide scale.
- It diagnoses and corrects econometric bias in fixed-effects estimates of fund-level scale.
- It measures skill as gross alpha before erosion by fund-level and industry-level scale.
- It explains why fund skill can rise while observed fund performance does not.
- It links the age-performance relation to industry growth rather than simple skill decay.

3 Data and measurement

The sample combines CRSP and Morningstar active domestic equity mutual funds from 1979–2011. The main sample begins in March 1993 because pre-1993 expense ratio, benchmark, and AUM data are less reliable. GrossR is benchmark-adjusted gross return. FundSize is lagged fund AUM scaled to 2011 market size. IndustrySize is total active equity mutual fund AUM divided by total CRSP stock-market capitalization.

Interpretation. Gross returns measure manager skill before fees; net returns measure investor performance. FundSize captures own trading capacity, while IndustrySize captures crowding by the active-management industry.

4 Models and empirical specifications

4.1 Fund-level scale

$$GrossR_{i,t} = a_i + \beta FundSize_{i,t-1} + \varepsilon_{i,t}.$$

Interpretation. A negative β means that a fund performs worse when its own size is larger. OLS and fixed effects are biased because fund size is endogenous to skill and past returns.

4.2 Recursive demeaning

$$R_{i,t} = a_i + \beta' x_{i,t-1} + \varepsilon_{i,t}.$$

Backward demean regressors and forward demean returns to remove fixed effects without the standard dynamic panel bias.

Interpretation. This estimator provides bias-free evidence about fund-level returns to scale, though with lower statistical power.

4.3 Industry-level scale

$$GrossR_{i,t} = a_i + \beta_I IndustrySize_{t-1} + \varepsilon_{i,t}.$$

Interpretation. A negative β_I means industry crowding lowers fund performance even for a given fund.

4.4 Heterogeneity

$$GrossR_{i,t} = a_i + \beta_I IndustrySize_{t-1} + \delta IndustrySize_{t-1} \times Characteristic_i + \varepsilon_{i,t}.$$

Interpretation. Negative interactions with small-cap, volatility, and turnover indicate that decreasing returns are strongest where trading capacity is limited.

4.5 Fund age

$$GrossR_{i,t} = a_i + AgeEffects_{i,t} + Controls_{i,t} + \varepsilon_{i,t}.$$

Interpretation. Age effects test whether performance declines within a fund over its life; controlling for IndustrySize distinguishes lifecycle effects from industry growth.

5 Identification and empirical design

The main challenge is endogeneity of fund size. The paper uses simulations and recursive demeaning to correct the bias. IndustrySize provides stronger evidence because it is aggregate and less mechanically tied to individual fund residuals. Time trends remain a concern, so the paper decomposes IndustrySize and conducts robustness checks using sector size, family size, fund age, and business-cycle controls.

6 Tables and figures

6.1 Table 1: Simulation exercise comparing OLS, fixed effects, and recursive demeaning + Table 2: Summary statistics

Read: Table 1 shows why OLS and fixed effects can be biased and why recursive demeaning is used. Table 2 shows the scale and dispersion of the mutual fund sample.

Economic message. Correcting estimator bias is essential before interpreting fund-level size effects.

Table 1
Simulation exercise comparing ordinary least squares (OLS), OLS fixed effect (OLS FE), and recursive demeaning (RD) estimators.

We simulate ten thousand samples of mutual fund gross returns (R_t) and size (z_t). Simulated returns follow $R_t = \alpha_0 + \alpha_1 R_{t-1} + \alpha_2 z_t + \epsilon_t$, and size follows $z_t = \beta_0 + \beta_1 z_{t-1} + \beta_2 R_{t-1} + \eta_t$. In each sample, we estimate β using the OLS, OLS FE, and RD estimators. Panel A (B) shows means (medians) of the β estimates across the simulated samples. Panel C shows the fraction of samples in which we reject the null hypothesis $\beta = 0$.

$\beta (\times 10^3)$	OLS, no FE			OLS with FE			RD		
	$\gamma=0.8$	$\gamma=0.9$	$\gamma=1.0$	$\gamma=0.8$	$\gamma=0.9$	$\gamma=1.0$	$\gamma=0.8$	$\gamma=0.9$	$\gamma=1.0$
Panel A: Mean estimated $\beta (\times 10^3)$									
0	0.85	0.84	0.82	-0.39	-0.38	-0.37	0.00	0.00	0.00
-1	0.30	0.37	0.42	-1.69	-1.73	-1.77	-0.97	-0.87	-1.03
-3	-0.96	-0.78	-0.62	-4.17	-4.28	-4.37	-2.34	-3.00	-3.08
-10	-7.51	-7.38	-7.27	-10.81	-10.82	-10.83	-10.03	-10.02	-10.02
Panel B: Median estimated $\beta (\times 10^3)$									
0	0.85	0.84	0.82	-0.39	-0.38	-0.37	0.00	0.00	0.00
-1	0.30	0.37	0.42	-1.68	-1.73	-1.76	-1.02	-1.01	-1.01
-3	-0.96	-0.78	-0.62	-4.17	-4.27	-4.37	-2.95	-2.95	-2.95
-10	-7.51	-7.38	-7.27	-10.81	-10.82	-10.83	-10.01	-10.01	-10.01
Panel C: Fraction rejecting the null hypothesis ($\beta=0$)									
0	1.00	1.00	1.00	0.98	0.99	0.99	0.06	0.06	0.06
-1	0.77	0.90	0.96	1.00	1.00	1.00	0.13	0.14	0.16
-3	1.00	0.99	0.94	1.00	1.00	1.00	0.21	0.20	0.19
-10	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

(a) Table 1: estimator simulation

Table 2
Summary statistics.

This table shows summary statistics for our sample of active equity mutual funds from 1979 to 2011. The unit of observation is the fund-month. All returns and expense ratios are in units of fraction per month. Net return is the return received by investors. Benchmark-adjusted net return equals net return minus the return on Morningstar's chosen benchmark portfolio. GrossR is the benchmark-adjusted net return plus 1/12th of the annual expense ratio. FundSize is the fund's total assets under management (AUM) aggregated across share classes, divided by the total stock market capitalization in the same month, then multiplied by the total stock market capitalization at the end of 2011. IndustrySize is the sum of all funds' AUM divided by the total stock market capitalization in the same month, imputing FundSize when missing. Turnover is in units of fraction per year. 150mCap is an indicator for a small-cap fund, as defined by Morningstar's Category variable. Std(AbRet) is the standard deviation of a fund's abnormal returns, defined as residuals from a regression of excess gross fund returns on excess benchmark portfolio returns. FundAge is the number of years since the fund's first offer date. FamilySize is the sum of FundSize across funds in the same family. SectorSize is the sum of AUM (in nominal dollars) across all funds within a given sector, divided by the total market value of all Center for Research in Security Prices stocks belonging to that sector. The first version uses the nine sectors in Morningstar's 3 x 3 Stylebox. The second version uses three sectors: small-cap, mid-cap, and large-cap.

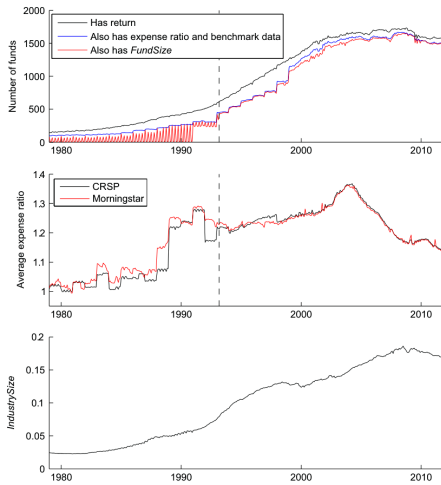
Variable	Number of fund-months	Mean	Standard deviation	Percentile				
				1st	25th	50th	75th	99th
Net return	365,182	0.0069	0.0529	-0.1518	-0.0197	0.0103	0.0371	0.1320
Benchmark-adjusted net return	345,921	-0.0005	0.0234	-0.0642	-0.0106	-0.0009	0.0091	0.0680
Benchmark-adjusted gross return (GrossR)	314,580	0.0005	0.0229	-0.0627	-0.0095	0.0001	0.0100	0.0680
Expense ratio	334,767	0.0010	0.0004	0.0003	0.0008	0.0010	0.0012	0.0021
FundSize (in millions of 2011 dollars)	319,454	1,564	5,779	16	84	265	921	24,371
IndustrySize	381,192	0.1134	0.0441	0.0232	0.1227	0.1391	0.1714	0.1833
Turnover	364,957	0.4478	0.5674	0.0660	0.4471	0.7394	1.0919	2.8521
150mCap	340,061	0.9319	0.3954	0	0	0	0	1
Std(AbRet)	367,495	0.0183	0.0084	0.0056	0.0125	0.0171	0.0222	0.0473
FundAge (years)	372,876	13.16	14.27	0.17	3.92	8.58	16.17	69.00
FamilySize (in millions of 2011 dollars)	315,894	61,821	149,667	23	1,768	12,181	46,217	825,507
SectorSize (version 1)	339,762	0.1561	0.0989	0.0220	0.0908	0.1290	0.2004	0.4788
SectorSize (version 2)	339,762	0.1390	0.0681	0.0210	0.1106	0.1318	0.1564	0.3550

(b) Table 2: summary statistics

6.2 Figure 1: Sample properties + Table 3: Relation between size and fund performance

Read: Figure 1 shows the increase in the number of funds and IndustrySize. Table 3 shows that industry-level scale strongly predicts lower GrossR, while bias-free fund-level scale estimates are weaker.

Economic message. Industry-level crowding is the paper's core empirical finding.



Sample properties. The top panel shows how the number of funds with non-missing observations changes over time. The blue line plots the number of funds that also have a non-missing cross-checked net return. The blue line plots the number of funds that also have a non-missing CRSP expense ratio and Morningstar benchmark return. The red line plots the number of funds that also have a non-missing FundSize. The vertical dashed line in the top two panels marks March 1993. The bottom panel shows the time series of the average expense ratio, in percent per year, for funds that have expense ratios from both sources. The bottom panel shows IndustrySize under management divided by the total market value of CRSP stocks. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this paper.)

out -9. However, this estimated relation could be indicate a negative relation between a fund

(a) Figure 1: sample properties

Table 3
Relation between size and fund performance.

The dependent variable in each regression model is GrossR, the fund's benchmark-adjusted gross return. FundSize is the fund's total assets under management (AUM) at the end of the previous month, inflated to millions of 2011 dollars using the total market cap of stocks in the Center for Research in Security Prices (CRSP) database. IndustrySize is the total AUM of all active equity mutual funds divided by the total market cap of all stocks in CRSP. The ordinary least squares fixed effect (OLS FE) estimator includes fund fixed effects. The recursive demeaning (RD) estimator recursively forward-demeans all variables and instruments for forward-demeaned FundSize using backward-demeaned FundSize. We multiply the slopes on FundSize by 10⁶ to make them easier to read. The reported slopes on FundSize thus equal the change in GrossR, in units of basis points per month, associated with a \$100 million increase in FundSize. Heteroskedasticity-robust t-statistics clustered by sector x month are in parentheses. We also cluster by fund in the RD specifications.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Main sample (March 1993–December 2011)									
FundSize	-0.0137 (-1.87)	-0.168 (-9.38)	-0.220 (-0.62)				-0.0147 (-2.02)	-0.148 (-9.09)	-0.425 (-1.25)
IndustrySize				-0.0169 (-1.93)	-0.0326 (-3.60)	-0.0326 (-2.49)	-0.0165 (-1.90)	-0.0295 (-3.27)	-0.0277 (-2.14)
Constant	0.000503 (2.18)			0.00304 (2.10)			0.00300 (2.09)		
Number of observations	275,847	275,847	270,556	283,046	283,046	283,046	275,847	270,556	
Estimator	OLS, no FE	OLS FE	RD	OLS, no FE	OLS FE	RD	OLS, no FE	OLS FE	RD
Panel B: Extended sample (January 1979–December 2011)									
FundSize	-0.0139 (-1.98)	-0.126 (-9.09)	-0.107 (-0.64)				-0.0161 (-2.30)	-0.103 (-7.91)	-0.268 (-1.69)
IndustrySize				-0.00671 (-1.92)	-0.0164 (-4.34)	-0.0165 (-2.66)	-0.00943 (-1.86)	-0.0165 (-3.15)	-0.0179 (-2.09)
Constant	0.000506 (2.28)			0.00144 (2.58)			0.00189 (2.26)		
Number of observations	290,150	290,150	285,350	314,580	314,580	314,580	290,150	285,350	
Estimator	OLS, no FE	OLS FE	RD	OLS, no FE	OLS FE	RD	OLS, no FE	OLS FE	RD

(b) Table 3: size-performance regressions

6.3 Table 4: Properties of industry-size effects + Table 5: Determinants in the main sample

Read: Table 4 shows IndustrySize remains negative when decomposed into time trend, average fund size, and number of funds. Table 5 shows decreasing returns are stronger for capacity-constrained strategies.

Economic message. Scale effects are not just a time trend; they are stronger in small-cap, volatile, and high-turnover strategies.

Table 4

Properties of the relation between industry size and fund performance.

The dependent variable in all regressions is *GrossR*, the fund's benchmark-adjusted gross return. *Time Trend* is the number of months since January 1979. *IndustrySize* is the total assets under management (AUM) of all sample funds divided by the total market cap of all stocks in the Center for Research in Security Prices database. *Average Fund Size* is the average AUM across all active equity mutual funds in a given month. We inflate *Average Fund Size* to current dollars by dividing by the total stock market capitalization in the same month, then multiplying by the total stock market capitalization at the end of 2011. *Number of Funds* is a count of the sample funds operating in the given month. *IndustrySize* equals *Number of Funds* times *Average Fund Size* divided by the total stock market capitalization at the end of 2011. We multiply the slopes on *Time Trend*, *Average Fund Size*, and *Number of Funds* by 10⁷ to make them easier to read. All models include fund fixed effects and are estimated by ordinary least squares. Heteroskedasticity-robust *t*-statistics clustered by sector $\times$ month are in parentheses.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Main sample (March 1993 – December 2011)								
<i>IndustrySize</i>	-0.0326 (-3.60)		-0.0852 (-3.04)			-0.115 (-2.60)	-0.146 (-2.87)	
<i>Time Trend</i>		-10.26 (-2.99)	23.89 (2.21)				15.35 (1.56)	
<i>Average Fund Size</i>				-3.862 (-3.03)	-8.885 (-3.56)	4.315 (0.73)	4.249 (0.72)	
<i>Number of Funds</i>					0.450 (0.83)	-4.031 (-3.23)	8.493 (1.61)	8.234 (1.57)
<i>Number of observations</i>	283,046	283,046	283,046	283,046	283,046	283,046	283,046	283,046
Panel B: Extended sample (January 1979 – December 2011)								
<i>IndustrySize</i>	-0.0164 (-4.34)		-0.0305 (-1.72)			-0.0629 (-1.93)	-0.0697 (-1.87)	
<i>Time Trend</i>		-7.030 (-4.00)	7.002 (0.84)				4.137 (0.51)	
<i>Average Fund Size</i>				-2.532 (-2.44)	-8.898 (-3.83)	-1.455 (-0.20)	-1.608 (-0.34)	
<i>Number of Funds</i>					0.0184 (0.05)	-3.026 (-4.26)	3.516 (0.84)	3.352 (0.81)
<i>Number of observations</i>	314,580	314,580	314,580	314,580	314,580	314,580	314,580	314,580

(a) Table 4: industry-size properties

Table 5

Determinants of the size-performance relation, main sample (March 1993–December 2011).

This table adds regressors to the specifications in Table 3. *1(SmCap)* is an indicator for whether the fund is a small-cap fund. *Std(AbRet)* is the fund's standard deviation of residuals from a regression of excess gross returns (in fraction per month) on excess benchmark returns. *Turnover* is the fund's average annual portfolio turnover. We instrument for all forward-demeaned quantities that involve *FundSize* by using the backward-demeaned values. We multiply the slope on *FundAge* by 1,000 to make it easier to read.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>FundSize</i>	-0.238 (-0.74)	-0.111 (-0.18)	0.0405 (0.08)				0.0881 (0.15)	0.206 (0.34)
<i>FundSize*1(SmCap)</i>	1.304 (0.42)						-0.660 (-0.23)	0.249 (0.09)
<i>FundSize*Std(AbRet)</i>		-11.41 (-0.30)					-28.69 (-0.83)	-31.18 (-0.89)
<i>FundSize*Turnover</i>			-0.319 (-0.61)				0.0106 (0.01)	-0.00319 (-0.01)
<i>IndustrySize</i>				-0.0242 (-2.34)	0.0547 (3.36)	-0.000687 (-0.06)	0.0617 (2.41)	0.0109 (0.30)
<i>IndustrySize*1(SmCap)</i>				-0.0523 (-2.09)			-0.0342 (-1.03)	-0.0367 (-1.11)
<i>IndustrySize*Std(AbRet)</i>					-4.533 (-4.80)		-3.476 (-2.82)	-3.529 (-2.97)
<i>IndustrySize*Turnover</i>						-0.0391 (-3.10)	-0.0218 (-1.65)	-0.0199 (-1.51)
<i>FundAge</i>								0.271 (1.97)
<i>Number of observations</i>	246,841	257,658	270,553	260,411	278,694	283,039	236,333	236,333

(b) Table 5: main-sample determinants

6.4 Table 6: Extended-sample determinants + Figure 2: Distribution of fund skill over time

Read: Table 6 confirms the heterogeneity results in the extended sample. Figure 2 shows fund skill rising over time.

Economic message. Skill improves over time, but the industry becomes too crowded for this to improve realized performance.

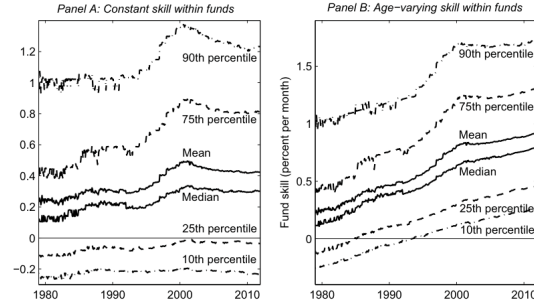
Table 6

Determinants of the size-performance relation, extended sample (January 1979–December 2011).

This table is the same as Table 5 but uses data from January 1979 to December 2011.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>FundSize</i>	-0.0987 (-0.66)	0.0228 (0.03)	-0.316 (-0.30)				0.271 (0.42)	0.318 (0.49)
<i>FundSize*1(SmCap)</i>	0.273 (0.13)						-1.402 (-0.70)	-0.959 (-0.49)
<i>FundSize*Std(AbRet)</i>		-10.40 (-0.28)					-29.83 (-0.94)	-30.19 (-0.94)
<i>FundSize*Turnover</i>			0.207 (0.21)				0.0588 (0.20)	0.0390 (0.12)
<i>IndustrySize</i>				-0.0120 (-3.04)	0.0248 (2.92)	0.00541 (1.11)	0.0450 (2.35)	0.0194 (0.68)
<i>IndustrySize*1(SmCap)</i>				-0.0348 (-2.67)			-0.0340 (-1.33)	-0.0360 (-1.41)
<i>IndustrySize*Std(AbRet)</i>					-2.137 (-4.51)		-2.013 (-2.19)	-2.010 (-2.19)
<i>IndustrySize*Turnover</i>						-0.0287 (-4.45)	-0.0250 (-2.57)	-0.0249 (-2.56)
<i>FundAge</i>								0.151 (1.23)
<i>Number of observations</i>	261,349	273,881	285,347	291,869	310,228	314,573	252,292	252,292

(a) Table 6: extended-sample determinants



1 skill over time. The figure plots each month's mean and percentiles of estimated fund skill across all funds operated by fund fixed effects estimated from the specification in Column 7 of Table 6. Panel B measures skill by α_i . $b = 0.000151$ are estimated from the specification in Column 8 of Table 6.

(b) Figure 2: skill distribution

6.5 Figure 3: Average fund returns over time + Figure 4: Time-series relation between fund age and fund performance

Read: Figure 3 shows no upward trend in realized returns, but the industry-size-adjusted line rises. Figure 4 shows performance declining with fund age.

Economic message. Rising skill is offset by industry growth; age-performance decline reflects funds aging in a more crowded industry.

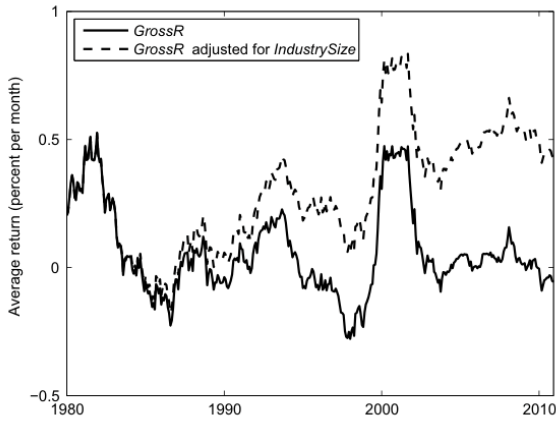


Fig. 3. Average fund returns over time. The figure plots two-year moving averages of fund returns. The solid line shows equal-weighted average benchmark-adjusted gross returns (*GrossR*). The dashed line is the same as the solid line except that it adjusts funds' *GrossR* by adding 0.0326 times ($IndustrySize_t - IndustrySize_0$), where $t=0$ denotes January 1979, the beginning of our sample. The value 0.0326 is minus the estimated slope from an ordinary least squares fixed effects regression of *GrossR* on fund fixed effects and *IndustrySize* (Table 3, Panel A, Column 5).

industry are less skilled, on average. However, this exit-based effect is unlikely to be the leading explanation of the growth in average skill because fund entry has far

(a) Figure 3: average returns over time

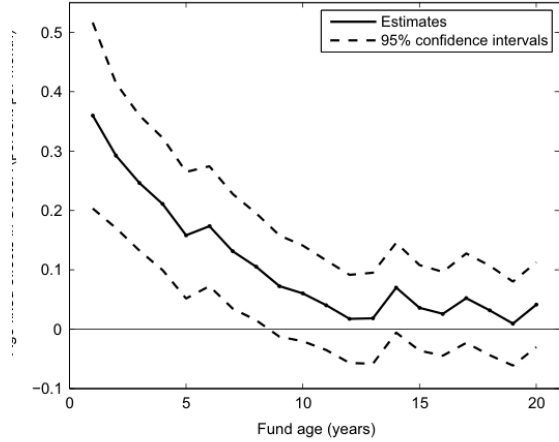


Fig. 4. Time series relation between fund age and fund performance. The figure plots fund age fixed effects in benchmark-adjusted gross returns (*GrossR*) along with their 95% confidence intervals. We obtain these fixed effects from a panel regression of *GrossR* on fund fixed effects and dummy variables for fund age = 1, 2, ..., 20 years. The age t fixed effect measures the average difference between *GrossR* for a fund of age t and the same fund at age 21+ years. Standard errors are clustered by sector $\times$ month. The sample covers March 1993 to December 2011.

otted). All of this evidence makes the incubation bias an unlikely explanation for our fund age result.

(b) Figure 4: fund age and performance

6.6 Table 7: Time-series relation between fund age and performance + Table 8: Investment strategies based on fund age

Read: Table 7 shows age is negative without *IndustrySize* but turns positive or insignificant once *IndustrySize* is controlled for. Table 8 shows young funds outperform older funds in age-sorted portfolios.

Economic message. The lifecycle evidence supports the industry-scale explanation rather than simple within-fund skill decay.

Table 7
Time-series relation between fund age and fund performance.
This table shows results from estimating a panel model with dependent variable *GrossR*, the fund's benchmark-adjusted gross return. *FundAge* is the number of years since the fund's first offer date. *IndustrySize* is the total assets under management (AUM) of all active equity mutual funds divided by the total market cap of all stocks in the Center for Research in Security Prices (CRSP) database. *FundSize* is the fund's total AUM at the end of the previous month, inflated to millions of 2011 dollars using the total market cap of stocks in CRSP. Specifications that do not include *FundSize* use the ordinary least squares fixed effect (OLS FE) estimator; those that do include *FundSize* use the recursive demeaning (RD) estimator. The first three columns in each panel use all observations; the last three columns use only funds that are at least three years old. We multiply the slopes on *FundAge* by 1,000 to make them easier to read. t -statistics clustered by sector $\times$ month are in parentheses.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Main sample (March 1993–December 2011)						
<i>FundAge</i>	−0.123 (−3.00)	0.283 (2.19)	0.277 (2.10)	−0.102 (−2.37)	0.281 (2.19)	0.267 (2.02)
<i>IndustrySize</i>		−0.0845 (−3.02)	−0.0794 (−2.77)	−0.0799 (−2.86)	−0.0746 (−2.60)	−0.0746 (−2.60)
<i>FundSize</i>			−0.264 (−0.78)		−0.259 (−0.77)	−0.204 (−0.77)
Number of observations	283,046	283,046	270,556	248,050	248,050	238,426
Fund ages	All	All	All	≥ 3 years	≥ 3 years	≥ 3 years
Panel B: Extended sample (January 1979–December 2011)						
<i>FundAge</i>	−0.0845 (−4.00)	0.0832 (0.82)	0.152 (1.28)	−0.0729 (−3.43)	0.0934 (0.93)	0.153 (1.29)
<i>IndustrySize</i>		−0.0302 (−1.70)	−0.0440 (−2.11)	−0.0298 (−1.69)	−0.0417 (−2.01)	−0.0417 (−2.01)
<i>FundSize</i>			−0.220 (−1.48)		−0.204 (−1.39)	−0.204 (−1.39)
Number of observations	314,580	314,580	285,350	277,381	277,381	252,222
Fund ages	All	All	All	≥ 3 years	≥ 3 years	≥ 3 years

(a) Table 7: fund age and performance

Table 8
Investment strategies based on fund age.
This table shows average returns of portfolios containing mutual funds of different ages. *FundAge* is the number of years since the fund's first offer date. At the beginning of each month, we assign mutual funds to portfolios based on their age at the end of the coming month. Columns 2–5 show the portfolios' average equal-weighted benchmark-adjusted gross (*GrossR*) and net (*NetR*) returns, in percent per month. The next three columns show the average difference in benchmark-adjusted returns between portfolios. The last column contains the p -value from an F -test of whether the four age-sorted portfolios have the same average benchmark-adjusted return, clustering by calendar date.

	Average portfolio return for fund age				Average differences for fund age				F-test
	[0, 3]	(3, 6]	(6, 10]	> 10	[0, 3] - (> 10)	(3, 6] - (> 10)	(6, 10] - (> 10)	p-value	
Panel A: Main sample (March 1993 – December 2011)									
Average <i>GrossR</i>	0.084 (2.33)	0.056 (1.45)	0.020 (0.55)	0.012 (0.30)	0.072 (2.85)	0.043 (2.48)	0.008 (0.52)	0.014	
Average <i>NetR</i>	-0.005 (-0.15)	-0.052 (-1.38)	-0.084 (-2.29)	-0.083 (-2.07)	0.077 (3.10)	0.031 (1.79)	-0.001 (-0.08)	0.008	
Panel B: Extended sample (January 1979 – December 2011)									
Average <i>GrossR</i>	0.106 (2.14)	0.120 (2.58)	0.105 (2.41)	0.033 (1.03)	0.075 (1.77)	0.096 (2.75)	0.072 (2.63)	0.003	
Average <i>NetR</i>	0.007 (0.18)	-0.018 (-0.46)	-0.033 (-0.86)	-0.057 (-1.78)	0.064 (2.69)	0.048 (1.77)	0.024 (1.06)	0.032	

(b) Table 8: age-based portfolios

7 Short summary

- Fund-level decreasing returns are difficult to estimate because fund size is endogenous.
- Recursive demeaning points to decreasing returns but lacks statistical significance.
- Industry-level decreasing returns are strong and robust.
- Skill rises over time, especially in the upper tail.
- Realized performance does not improve because rising IndustrySize offsets rising skill.
- Fund performance declines over the lifecycle, but this is explained by industry growth.

8 Conclusion

8.1 Core conclusion

Active management faces decreasing returns to scale, especially at the industry level.

8.2 Incremental contribution

The paper connects scale, skill, and lifecycle performance in one framework and shows that skill can rise while returns stagnate because industry-wide crowding erodes alpha.

8.3 Economic intuition

Mispricing opportunities are finite. As more assets chase them, price impact increases and alpha falls.

8.4 Implications

Observed alpha is not a clean measure of skill unless one adjusts for fund-level and industry-level scale. Investors and researchers should distinguish manager ability from the opportunity set facing the manager.

8.5 Limitations and future work

IndustrySize is observational and trends upward over time; the exact structural channel behind decreasing returns remains open for future research.